TFM

Importación de librerías

```
!pip install --upgrade tensorflow
```

```
!pip show tensorflow
!pip show keras
```

```
import pandas as pd
import numpy as np
import os
from glob import glob
import itertools
import fnmatch
import random
import math
import matplotlib.pylab as plt
import seaborn as sns
import cv2
import sklearn
import keras
import tensorflow as tf
```

```
from sklearn.model_selection import train_test_split
from tensorflow.keras.preprocessing.image import ImageDataGenerator
from tensorflow.keras.preprocessing.image import load_img, img_to_array

from tensorflow.keras.applications import VGG16
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense, Flatten

from tensorflow.keras.applications import Xception
from tensorflow.keras.layers import Dense, GlobalAveragePooling2D

from tensorflow.keras.callbacks import ModelCheckpoint
from tensorflow.keras.callbacks import EarlyStopping

from sklearn.metrics import classification_report, accuracy_score, confusion_matrix, ConfusionMatrixDisplay
```

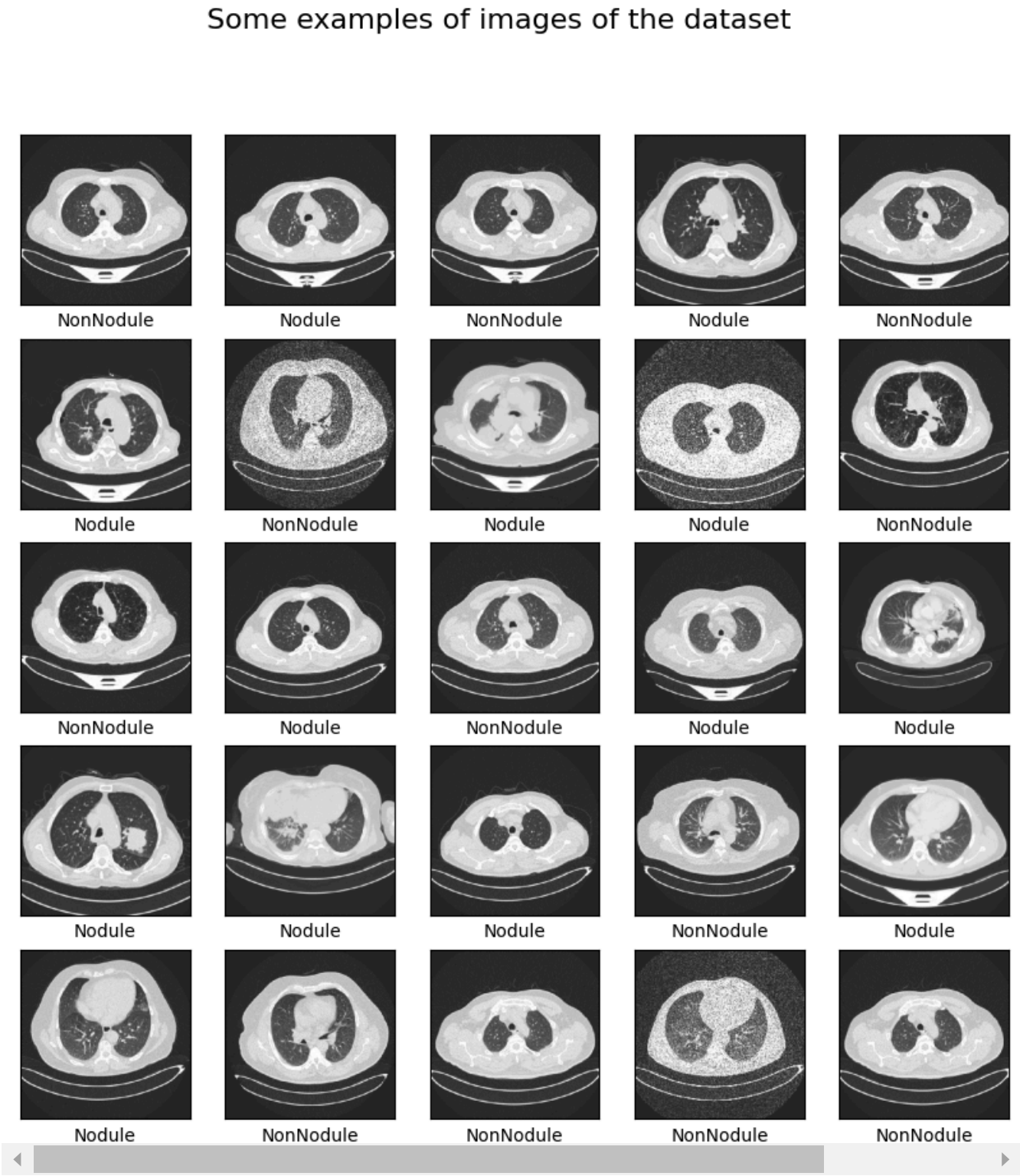
Preparación del dataset

```
# Rutas a las carpetas
data_dir = '/content/dataset/'

# Lista de todas las clases
clases = os.listdir(data_dir)
clases.remove('.ipynb_checkpoints')

# Listas para almacenar las rutas de las imágenes y las etiquetas
image_paths = []
labels = []
```

```
# Recorrer cada clase y agregar las rutas de las imágenes y etiquetas correspondientes
for clase in clases:
    clase_path = os.path.join(data_dir, clase)
    images = os.listdir(clase_path)
    image_paths += [os.path.join(clase_path, img) for img in images]
    labels += [clase] * len(images)


# Convertir etiquetas a números (opcional)
label_to_index = {label: idx for idx, label in enumerate(clases)}
numeric_labels = [label_to_index[label] for label in labels]

# Dividir en entrenamiento y validación (80% - 20%)
train_paths, val_paths, train_labels, val_labels = train_test_split(image_paths, numeric_labels, test_size=

# Crear generadores de datos personalizados a partir de las rutas
train_datagen = ImageDataGenerator(rescale=1.0/255)
val_datagen = ImageDataGenerator(rescale=1.0/255)

def custom_generator(paths, labels, batch_size, datagen):
    while True:
        # Mezclar los datos en cada iteración
        indices = np.arange(len(paths))
        np.random.shuffle(indices)
        for i in range(0, len(paths), batch_size):
            batch_indices = indices[i:i + batch_size]
            batch_paths = [paths[j] for j in batch_indices]
            batch_labels = [labels[j] for j in batch_indices]
            # batch_images = [datagen.flow_from_directory(batch_paths)]

            batch_images = []
            for path in batch_paths:
                img = load_img(path, target_size=(224, 224))  # Ajusta el tamaño según tu modelo
                img_array = img_to_array(img)
                img_array = datagen.random_transform(img_array)  # Aplicar aumentaciones
                img_array = datagen.standardize(img_array)  # Aplicar normalización (rescale)
                batch_images.append(img_array)

            yield np.array(batch_images), np.array(batch_labels)

train_generator = custom_generator(train_paths, train_labels, batch_size=32, datagen=train_datagen)
val_generator = custom_generator(val_paths, val_labels, batch_size=32, datagen=val_datagen)


def display_examples(class_names, images, labels):
    """
        Display 25 images from the images array with its corresponding labels
    """

    fig = plt.figure(figsize=(10,10))
    fig.suptitle("Some examples of images of the dataset", fontsize=16)
    for i in range(25):
        plt.subplot(5,5,i+1)
        plt.xticks([])
        plt.yticks([])
        plt.grid(False)
        plt.imshow(images[i], cmap=plt.cm.binary)
        plt.xlabel(class_names[labels[i]])
    plt.show()


# Get one batch of data
images, labels = next(train_generator)

print(images.shape, labels.shape)
```

```
display_examples(clases, images, labels)
```

```
(32, 224, 224, 3) (32,)
```



Definir y entrenar el modelo CNN

```
# model = tf.keras.models.Sequential([
#     tf.keras.layers.Conv2D(32, (3, 3), activation='relu', input_shape=(150, 150, 3)),
#     tf.keras.layers.MaxPooling2D(2, 2),

#     tf.keras.layers.Conv2D(64, (3, 3), activation='relu'),
#     tf.keras.layers.MaxPooling2D(2, 2),

#     tf.keras.layers.Conv2D(128, (3, 3), activation='relu'),
```

```
#     tf.keras.layers.MaxPooling2D(2, 2),

#     tf.keras.layers.Flatten(),
#     tf.keras.layers.Dense(512, activation='relu'),
#     tf.keras.layers.Dense(1, activation='sigmoid')  # Para clasificación binaria, usa 'softmax' para múlt
# ])

# Definir la entrada
inputs = tf.keras.Input(shape=(224, 224, 3))

# Capas convolucionales con MaxPooling
x = tf.keras.layers.Conv2D(32, (3, 3), activation='relu')(inputs)
x = tf.keras.layers.MaxPooling2D((2, 2))(x)

x = tf.keras.layers.Conv2D(64, (3, 3), activation='relu')(x)
x = tf.keras.layers.MaxPooling2D((2, 2))(x)

x = tf.keras.layers.Conv2D(128, (3, 3), activation='relu')(x)
x = tf.keras.layers.MaxPooling2D((2, 2))(x)

# Aplanamos el resultado
x = tf.keras.layers.Flatten()(x)

# # Capa densa intermedia
x = tf.keras.layers.Dense(512, activation='relu')(x)

# Congelamos esta parte del modelo
# x.trainable = False

# Capa de salida, para clasificación binaria o multiclase
x = tf.keras.layers.Dense(2, activation='softmax')(x)  # Para binario, si es multiclase, usa softmax y ajus

# Definir el modelo
model = tf.keras.Model(inputs, x)

# Compilar el modelo
model.compile(optimizer='adam', loss='sparse_categorical_crossentropy', metrics=['accuracy'])

# Resumen del modelo
model.summary()
```

Model: "functional_2"

Layer (type)	Output Shape	Param #
input_layer_2 (InputLayer)	(None, 224, 224, 3)	0
conv2d_6 (Conv2D)	(None, 222, 222, 32)	896
[illegible]	[illegible]	0
conv2d_7 (Conv2D)	(None, 109, 109, 64)	18,496
max_pooling2d_7 (MaxPooling2D)	(None, 54, 54, 64)	0
conv2d_8 (Conv2D)	(None, 52, 52, 128)	73,856
max_pooling2d_8 (MaxPooling2D)	(None, 26, 26, 128)	0
flatten_2 (Flatten)	(None, 86528)	0
dense_4 (Dense)	(None, 512)	44,302,848
dense_5 (Dense)	(None, 2)	1,026

Total params: 44,397,122 (169.36 MB)

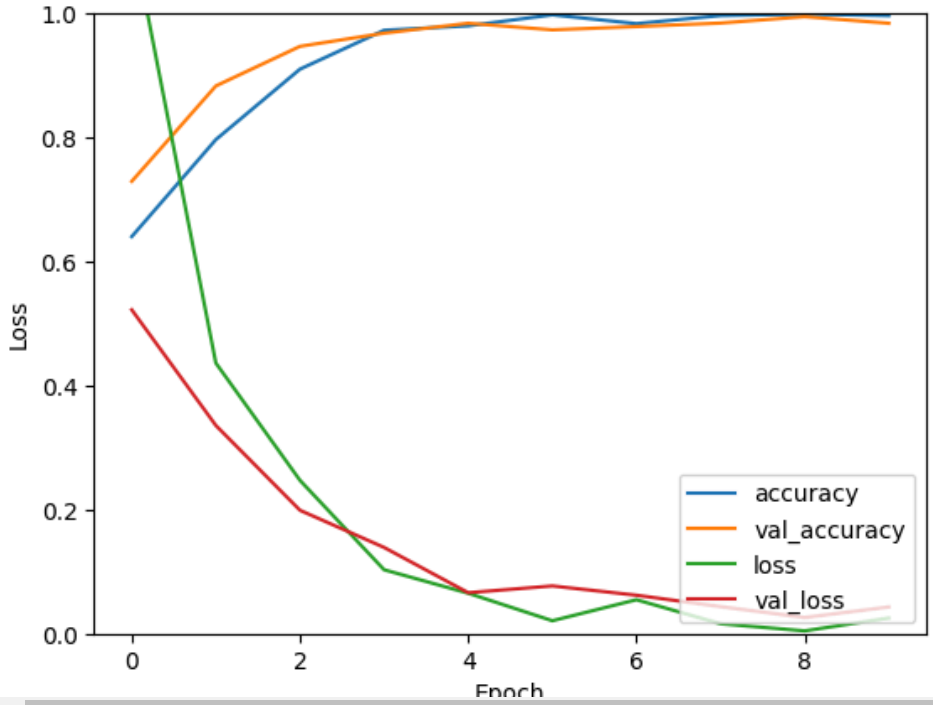
```
checkpoints = ModelCheckpoint('modelo.keras', monitor='val_loss', save_best_only=True)
early_stop = EarlyStopping(monitor='val_loss', patience=10)
callbacks_list = [checkpoints, early_stop]

history = model.fit(
    train_generator,
    steps_per_epoch=len(train_paths) // 32,
    validation_data=val_generator,
    validation_steps=len(val_paths) // 32,
    epochs=10,
    callbacks=callbacks_list,
    shuffle=True
)
```

```
Epoch 1/10
27/27 ━━━━━━━━━━━━━━━━━━━━ 137s 5s/step - accuracy: 0.6235 - loss: 2.0094 - val_accuracy: 0.7292 - val_
Epoch 2/10
27/27 ━━━━━━━━━━━━━━━━━━━━ 127s 5s/step - accuracy: 0.7706 - loss: 0.4656 - val_accuracy: 0.8830 - val_
Epoch 3/10
27/27 ━━━━━━━━━━━━━━━━━━━━ 123s 5s/step - accuracy: 0.9158 - loss: 0.2549 - val_accuracy: 0.9468 - val_
Epoch 4/10
27/27 ━━━━━━━━━━━━━━━━━━━━ 127s 5s/step - accuracy: 0.9694 - loss: 0.1162 - val_accuracy: 0.9681 - val_
Epoch 5/10
27/27 ━━━━━━━━━━━━━━━━━━━━ 127s 5s/step - accuracy: 0.9803 - loss: 0.0637 - val_accuracy: 0.9840 - val_
Epoch 6/10
27/27 ━━━━━━━━━━━━━━━━━━━━ 116s 4s/step - accuracy: 0.9972 - loss: 0.0290 - val_accuracy: 0.9734 - val_
Epoch 7/10
27/27 ━━━━━━━━━━━━━━━━━━━━ 145s 5s/step - accuracy: 0.9780 - loss: 0.0678 - val_accuracy: 0.9787 - val_
Epoch 8/10
27/27 ━━━━━━━━━━━━━━━━━━━━ 155s 6s/step - accuracy: 0.9960 - loss: 0.0191 - val_accuracy: 0.9844 - val_
Epoch 9/10
27/27 ━━━━━━━━━━━━━━━━━━━━ 144s 5s/step - accuracy: 0.9990 - loss: 0.0053 - val_accuracy: 0.9947 - val_
Epoch 10/10
27/27 ━━━━━━━━━━━━━━━━━━━━ 117s 4s/step - accuracy: 0.9975 - loss: 0.0102 - val_accuracy: 0.9840 - val_
```

Gráficos de evolución del modelo

```
# Accuracy
plt.plot(history.history['accuracy'], label='accuracy')
plt.plot(history.history['val_accuracy'], label = 'val_accuracy')
plt.xlabel('Epoch')
plt.ylabel('Accuracy')
plt.ylim([0, 1])
plt.legend(loc='lower right')

# Loss
plt.plot(history.history['loss'], label='loss')
plt.plot(history.history['val_loss'], label = 'val_loss')
plt.xlabel('Epoch')
plt.ylabel('Loss')
plt.ylim([0, 1])
plt.legend(loc='lower right')
```

```
<matplotlib.legend.Legend at 0x7a6c3720b010>
```



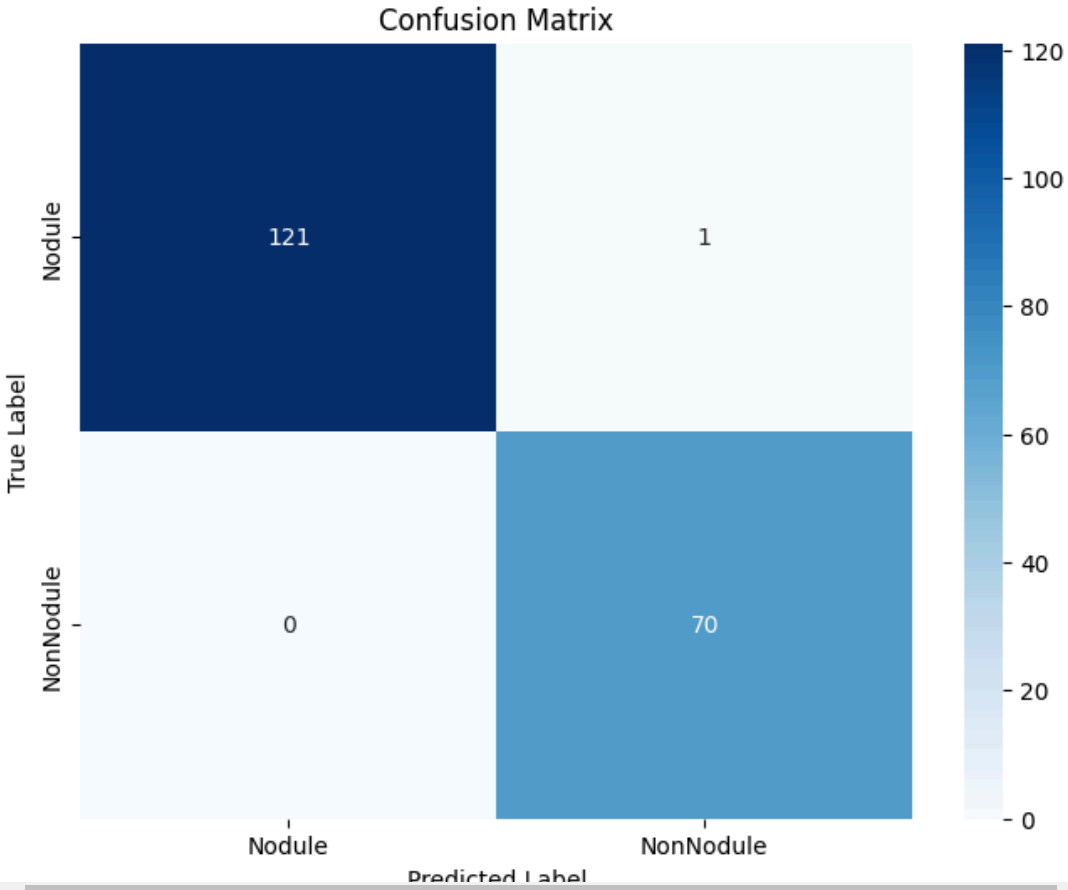
Evaluando el modelo

```
model = tf.keras.models.load_model('modelo.keras')
```

```
val_steps = len(val_paths) // 32

y_true = []
y_pred = []
for i in range(val_steps):
  images, labels = next(val_generator)
  predictions = model.predict(images, verbose=0)
  # predict_classes = (predictions > 0.5).astype("int32")
  predict_classes = np.argmax(predictions, axis=1)
  y_true.extend(labels)
  y_pred.extend(predict_classes)

# Convert to NumPy arrays for easier handling
y_true = np.array(y_true)
y_pred = np.array(y_pred)

# Calcular métricas de evaluación
precision = precision_score(y_true, y_pred, average='binary')  # 'micro' o 'macro' si tienes más clases
recall = recall_score(y_true, y_pred, average='binary')
f1 = f1_score(y_true, y_pred, average='binary')
accuracy = accuracy_score(y_true, y_pred)

# Imprimir resultados
print('')
print(f'Precision: {precision:.2f}')
print(f'Recall: {recall:.2f}')
print(f'F1 Score: {f1:.2f}')
print(f'Accuracy: {accuracy:.2f}')  # Imprimir exactitud
print('')

# Generar la matriz de confusión
cm = confusion_matrix(y_true, y_pred)
```

```
plt.figure(figsize=(8, 6))
# Replace val_generator.class_indices.keys() with your actual class labels
class_labels = ['Nodule', 'NonNodule'] # Update this with your actual class names
sns.heatmap(cm, annot=True, fmt="d", cmap="Blues", xticklabels=class_labels, yticklabels=class_labels)
plt.ylabel('True Label')
plt.xlabel('Predicted Label')
plt.title('Confusion Matrix')
plt.show()
```

```
Precision: 0.99
Recall: 1.00
F1 Score: 0.99
Accuracy: 0.99
```



Modelos Preentrenados de Keras y Evaluación

Modelo VGG16

```
# Cargar la arquitectura de VGG16 preentrenada (excluyendo la parte superior/fully connected)
base_model = VGG16(weights='imagenet', include_top=False, input_shape=(150, 150, 3))

# Congelar las capas de VGG16 para que no se entrenen
for layer in base_model.layers:
    layer.trainable = False

# Crear un nuevo modelo secuencial y añadir la base de VGG16
modelVGG = Sequential([
    base_model,
    Flatten(),
    Dense(512, activation='relu'),  # Añadir una capa densa (puedes ajustar el número de neuronas)
    Dense(1, activation='sigmoid')  # Capa de salida (ajusta para múltiples clases)
```

```
])

modelVGG.summary()


inputs = tf.keras.Input(shape=(224, 224, 3))

x = tf.keras.applications.vgg16.VGG16(
    input_tensor=inputs,
    include_top=False,
    weights='imagenet'
)

x.trainable = False

x = tf.keras.layers.Flatten()(x.output)
x = tf.keras.layers.Dense(2, activation='sigmoid')(x)

modelVGG = tf.keras.Model(inputs, x)

modelVGG.summary()
```

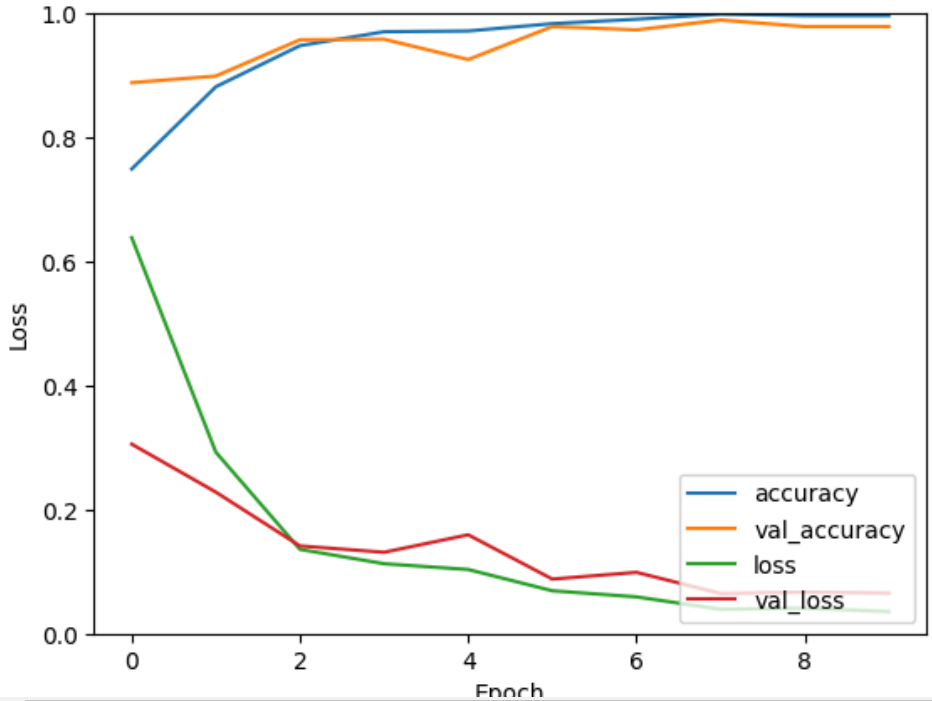
Mostrar salida oculta

```
#####
checkpoints = ModelCheckpoint('modeloVGG.keras', monitor='val_loss', save_best_only=True)
early_stop = EarlyStopping(monitor='val_loss', patience=10)
callbacks_list = [checkpoints, early_stop]
#####

# adam = tf.keras.optimizers.Adam(learning_rate=0.0001)
# model.compile(loss='categorical_crossentropy', metrics=['accuracy'], optimizer=adam)

# Compilar el modelo
modelVGG.compile(optimizer='adam', loss='sparse_categorical_crossentropy', metrics=['accuracy'])


history = modelVGG.fit(
    train_generator,
    steps_per_epoch=len(train_paths) // 32,  # Replace 32 with your actual batch size
    validation_data=val_generator,
    validation_steps=len(val_paths) // 32,  # Replace 32 with your actual batch size
    epochs=10,  # Adjust the number of epochs as needed
    callbacks=callbacks_list,
    shuffle=True
)
```

```
Epoch 1/10
27/27 ━━━━━━━━━━━━━━━━━━━━ 573s 21s/step - accuracy: 0.6746 - loss: 0.9132 - val_accuracy: 0.8883 - val
Epoch 2/10
27/27 ━━━━━━━━━━━━━━━━━━━━ 572s 21s/step - accuracy: 0.8615 - loss: 0.3222 - val_accuracy: 0.8989 - val
Epoch 3/10
27/27 ━━━━━━━━━━━━━━━━━━━━ 569s 21s/step - accuracy: 0.9384 - loss: 0.1545 - val_accuracy: 0.9574 - val
Epoch 4/10
27/27 ━━━━━━━━━━━━━━━━━━━━ 571s 21s/step - accuracy: 0.9649 - loss: 0.1234 - val_accuracy: 0.9583 - val
Epoch 5/10
27/27 ━━━━━━━━━━━━━━━━━━━━ 567s 21s/step - accuracy: 0.9744 - loss: 0.1055 - val_accuracy: 0.9255 - val
Epoch 6/10
27/27 ━━━━━━━━━━━━━━━━━━━━ 568s 21s/step - accuracy: 0.9795 - loss: 0.0722 - val_accuracy: 0.9787 - val
Epoch 7/10
27/27 ━━━━━━━━━━━━━━━━━━━━ 564s 21s/step - accuracy: 0.9885 - loss: 0.0653 - val_accuracy: 0.9734 - val
Epoch 8/10
27/27 ━━━━━━━━━━━━━━━━━━━━ 564s 21s/step - accuracy: 0.9992 - loss: 0.0384 - val_accuracy: 0.9894 - val
Epoch 9/10
27/27 ━━━━━━━━━━━━━━━━━━━━ 563s 21s/step - accuracy: 0.9963 - loss: 0.0423 - val_accuracy: 0.9787 - val
Epoch 10/10
27/27 ━━━━━━━━━━━━━━━━━━━━ 565s 21s/step - accuracy: 0.9966 - loss: 0.0346 - val_accuracy: 0.9787 - val
```

```
# Accuracy
plt.plot(history.history['accuracy'], label='accuracy')
plt.plot(history.history['val_accuracy'], label = 'val_accuracy')
plt.xlabel('Epoch')
plt.ylabel('Accuracy')
plt.ylim([0, 1])
plt.legend(loc='lower right')

# Loss
plt.plot(history.history['loss'], label='loss')
plt.plot(history.history['val_loss'], label = 'val_loss')
plt.xlabel('Epoch')
plt.ylabel('Loss')
plt.ylim([0, 1])
plt.legend(loc='lower right')
```

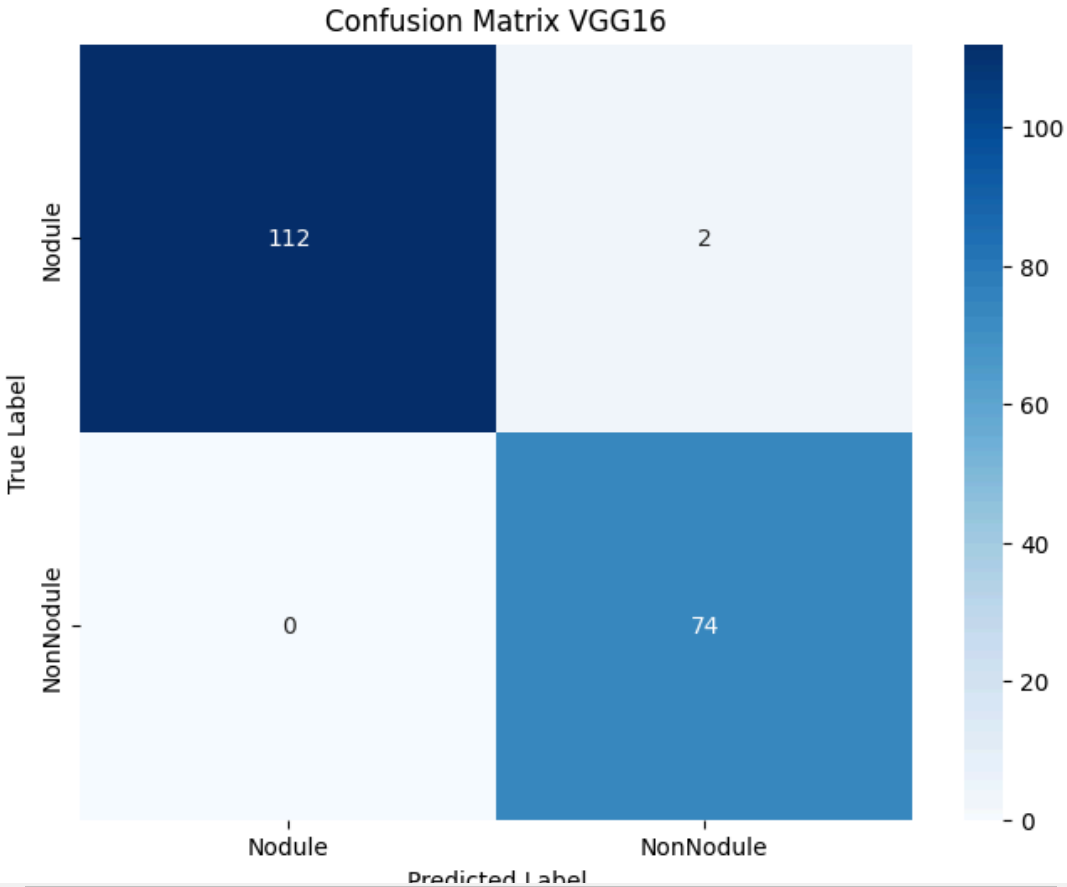
```
<matplotlib.legend.Legend at 0x7a6c3676f010>
```




```
modelVGG = tf.keras.models.load_model('modeloVGG.keras')


val_steps = len(val_paths) // 32  # Ajusta esto si usas otro tamaño de batch

y_true = []
y_pred = []
for i in range(val_steps):
  images, labels = next(val_generator)
  predictions = modelVGG.predict(images, verbose=0)
  # predict_classes = (predictions > 0.5).astype("int32")
  predict_classes = np.argmax(predictions, axis=1)

  y_true.extend(labels)
  y_pred.extend(predict_classes)

# Convert to NumPy arrays for easier handling
y_true = np.array(y_true)
y_pred = np.array(y_pred)

# Calcular métricas de evaluación
```

```
precision = precision_score(y_true, y_pred, average='binary')  # 'micro' o 'macro' si tienes más clases
recall = recall_score(y_true, y_pred, average='binary')
f1 = f1_score(y_true, y_pred, average='binary')
accuracy = accuracy_score(y_true, y_pred)
# Imprimir resultados
print('')
print(f'Precision: {precision:.2f}')
print(f'Recall: {recall:.2f}')
print(f'F1 Score: {f1:.2f}')
print(f'Accuracy: {accuracy:.2f}')  # Imprimir exactitud
print('')

cm = confusion_matrix(y_true, y_pred)

plt.figure(figsize=(8, 6))
# Replace val_generator.class_indices.keys() with your actual class labels
class_labels = ['Nodule', 'NonNodule'] # Update this with your actual class names
sns.heatmap(cm, annot=True, fmt="d", cmap="Blues", xticklabels=class_labels, yticklabels=class_labels)
plt.ylabel('True Label')
plt.xlabel('Predicted Label')
plt.title('Confusion Matrix VGG16')
plt.show()
```

```
Precision: 0.97
Recall: 1.00
F1 Score: 0.99
Accuracy: 0.99
```



Modelo Xception

```
inputs = tf.keras.Input(shape=(224, 224, 3))

x = tf.keras.applications.xception.Xception(
```

```
    input_tensor=inputs,
    include_top=False,
    weights='imagenet'
)

x.trainable = False

x = tf.keras.layers.Flatten()(x.output)
x = tf.keras.layers.Dense(2, activation='softmax')(x) # 1, sigmoid

modelXce = tf.keras.Model(inputs, x)

modelXce.summary()
```

Mostrar salida oculta

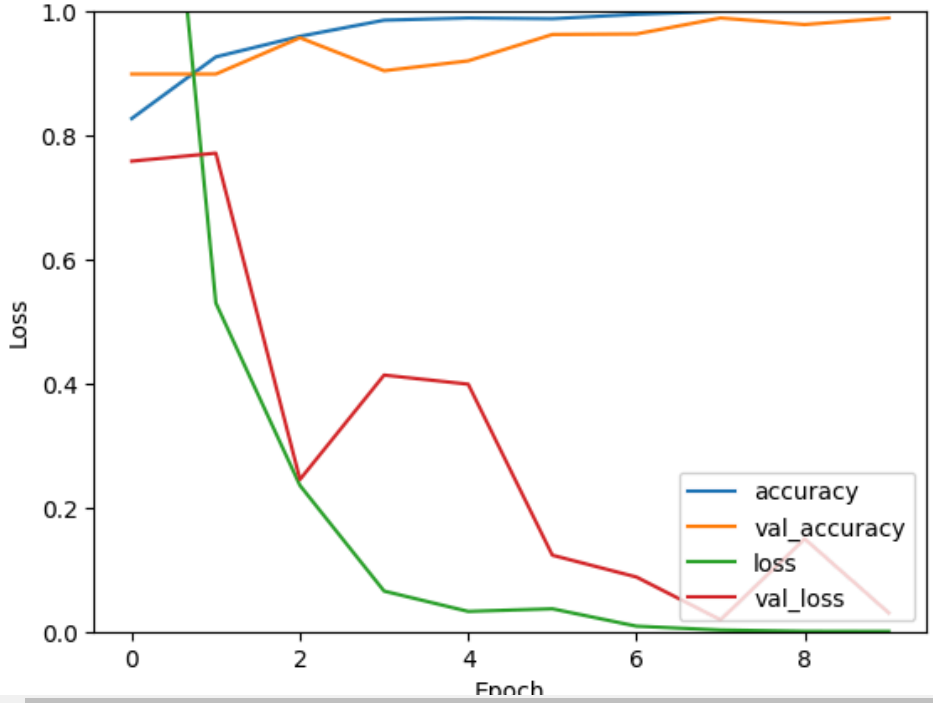
```
#####
checkpoints = ModelCheckpoint('modeloXce.keras', monitor='val_loss', save_best_only=True)
early_stop = EarlyStopping(monitor='val_loss', patience=10)
callbacks_list = [checkpoints, early_stop]
#####

# adam = tf.keras.optimizers.Adam(learning_rate=0.0001)
# model.compile(loss='categorical_crossentropy', metrics=['accuracy'], optimizer=adam)

# Compilar el modelo
modelXce.compile(optimizer='adam', loss='sparse_categorical_crossentropy', metrics=['accuracy']) # binary_c


history = modelXce.fit(
    train_generator,
    steps_per_epoch=len(train_paths) // 32,  # Replace 32 with your actual batch size
    validation_data=val_generator,
    validation_steps=len(val_paths) // 32,  # Replace 32 with your actual batch size
    epochs=10,  # Adjust the number of epochs as needed
    callbacks=callbacks_list,
    shuffle=True
)
```

```
Epoch 1/10
27/27 ━━━━━━━━━━━━━━━━━━━━ 238s 8s/step - accuracy: 0.7300 - loss: 2.5724 - val_accuracy: 0.8989 - val_
Epoch 2/10
27/27 ━━━━━━━━━━━━━━━━━━━━ 219s 8s/step - accuracy: 0.9104 - loss: 0.6111 - val_accuracy: 0.8989 - val_
Epoch 3/10
27/27 ━━━━━━━━━━━━━━━━━━━━ 217s 8s/step - accuracy: 0.9470 - loss: 0.3579 - val_accuracy: 0.9574 - val_
Epoch 4/10
27/27 ━━━━━━━━━━━━━━━━━━━━ 220s 8s/step - accuracy: 0.9843 - loss: 0.0732 - val_accuracy: 0.9043 - val_
Epoch 5/10
27/27 ━━━━━━━━━━━━━━━━━━━━ 217s 8s/step - accuracy: 0.9932 - loss: 0.0244 - val_accuracy: 0.9202 - val_
Epoch 6/10
27/27 ━━━━━━━━━━━━━━━━━━━━ 216s 8s/step - accuracy: 0.9870 - loss: 0.0732 - val_accuracy: 0.9628 - val_
Epoch 7/10
27/27 ━━━━━━━━━━━━━━━━━━━━ 218s 8s/step - accuracy: 0.9926 - loss: 0.0112 - val_accuracy: 0.9635 - val_
Epoch 8/10
27/27 ━━━━━━━━━━━━━━━━━━━━ 217s 8s/step - accuracy: 1.0000 - loss: 0.0033 - val_accuracy: 0.9894 - val_
Epoch 9/10
27/27 ━━━━━━━━━━━━━━━━━━━━ 219s 8s/step - accuracy: 1.0000 - loss: 3.6716e-04 - val_accuracy: 0.9787 -
Epoch 10/10
27/27 ━━━━━━━━━━━━━━━━━━━━ 219s 8s/step - accuracy: 1.0000 - loss: 2.2755e-04 - val_accuracy: 0.9894 -
```

```
# Accuracy
plt.plot(history.history['accuracy'], label='accuracy')
plt.plot(history.history['val_accuracy'], label = 'val_accuracy')
plt.xlabel('Epoch')
plt.ylabel('Accuracy')
```

```
plt.ylim([0, 1])
plt.legend(loc='lower right')

# Loss
plt.plot(history.history['loss'], label='loss')
plt.plot(history.history['val_loss'], label = 'val_loss')
plt.xlabel('Epoch')
plt.ylabel('Loss')
plt.ylim([0, 1])
plt.legend(loc='lower right')
```

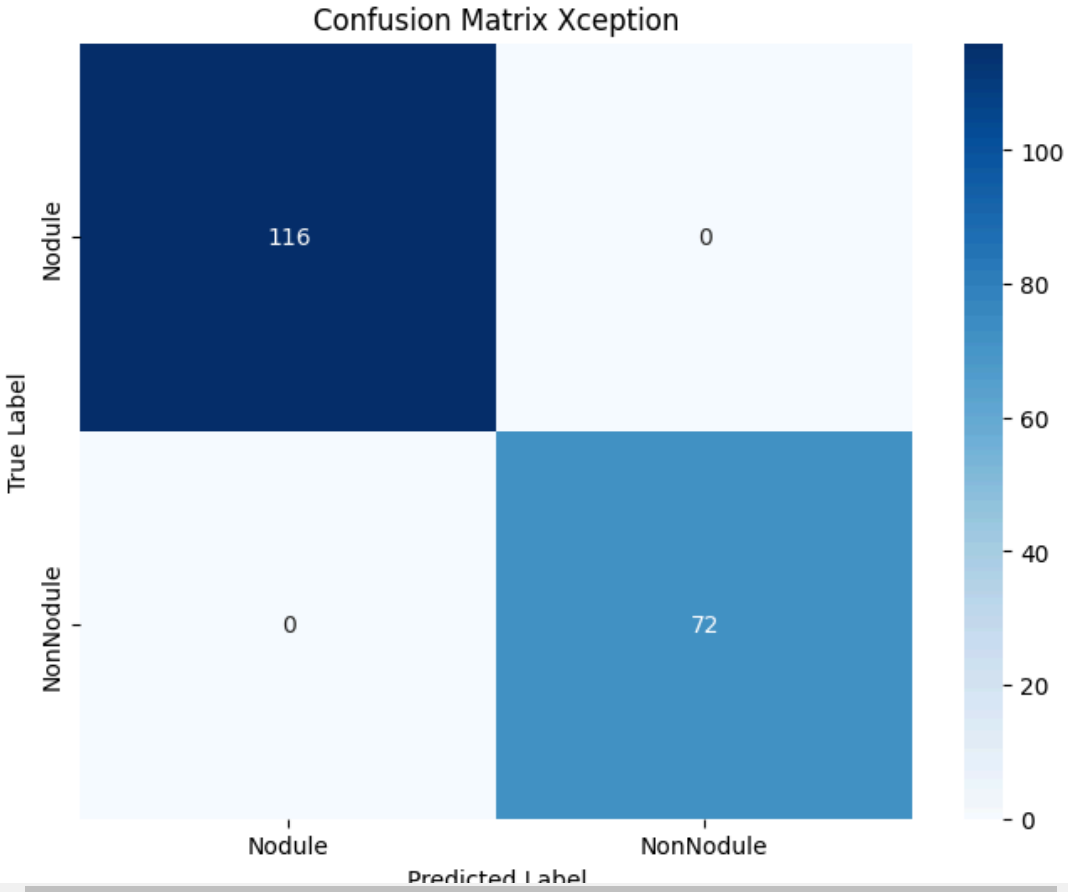
```
<matplotlib.legend.Legend at 0x7a6c5bbf8400>
```




```
modelXce = tf.keras.models.load_model('modeloXce.keras')


val_steps = len(val_paths) // 32  # Ajusta esto si usas otro tamaño de batch

y_true = []
y_pred = []
for i in range(val_steps):
  images, labels = next(val_generator)
  predictions = modelXce.predict(images, verbose=0)
  # predict_classes = (predictions > 0.5).astype("int32")
  # Use argmax to get the predicted class for binary classification
  predict_classes = np.argmax(predictions, axis=1)
  y_true.extend(labels)
  y_pred.extend(predict_classes)

# Convert to NumPy arrays for easier handling
y_true = np.array(y_true)
y_pred = np.array(y_pred)

# Calcular métricas de evaluación
precision = precision_score(y_true, y_pred, average='binary')  # 'micro' o 'macro' si tienes más clases
recall = recall_score(y_true, y_pred, average='binary')
f1 = f1_score(y_true, y_pred, average='binary')
accuracy = accuracy_score(y_true, y_pred)
# Imprimir resultados
print('')
print(f'Precision: {precision:.2f}')
print(f'Recall: {recall:.2f}')
```

```
print(f'F1 Score: {f1:.2f}')
print(f'Accuracy: {accuracy:.2f}')  # Imprimir exactitud
print('')

cm = confusion_matrix(y_true, y_pred)

plt.figure(figsize=(8, 6))
# Replace val_generator.class_indices.keys() with your actual class labels
class_labels = ['Nodule', 'NonNodule'] # Update this with your actual class names
sns.heatmap(cm, annot=True, fmt="d", cmap="Blues", xticklabels=class_labels, yticklabels=class_labels)
plt.ylabel('True Label')
plt.xlabel('Predicted Label')
plt.title('Confusion Matrix Xception')
plt.show()
```

```
Precision: 1.00
Recall: 1.00
F1 Score: 1.00
Accuracy: 1.00
```



Evaluación Modelos

```
model = tf.keras.models.load_model('modeloXce.keras')
```

```
model = tf.keras.models.load_model('modeloVGG.keras')
```

```
model = tf.keras.models.load_model('modelo.keras')
```

```
test_datagen = ImageDataGenerator(rescale=1.0/255)
```

```
test_generator = test_datagen.flow_from_directory(
    '/content/dataset/test',
    target_size=(224, 224),
    batch_size=32,
    class_mode='binary'
)


test_data = os.listdir('/content/test/images/')


def data_read_class(path):
  with open(path, 'r') as f:
    labels = f.read().split()
    class_label = labels[0]
  return class_label



TP = 0
TN = 0
FP = 0
FN = 0

for i in range(len(test_data)):
  img = load_img('/content/test/images/' + test_data[i], target_size=(224, 224))
  img_array = img_to_array(img)
  img_array = np.expand_dims(img_array, axis=0)
  img_array /= 255.0

  prediction = model.predict(img_array)

  if np.argmax(prediction[0]) > 0.5:
    predicted_class = 1
  else:
    predicted_class = 0

  if int(data_read_class(f'/content/test/labels/{test_data[i].replace(".jpg", ".txt")}')) == predicted_clas
    if predicted_class == 0: # La etiqueta 0 corresponde con la clase positiva -> Presencia de tumor
      TP += 1
    else:
      TN += 1
  else:
    if predicted_class == 0:
      FP += 1
    else:
      FN += 1
```

Mostrar salida oculta

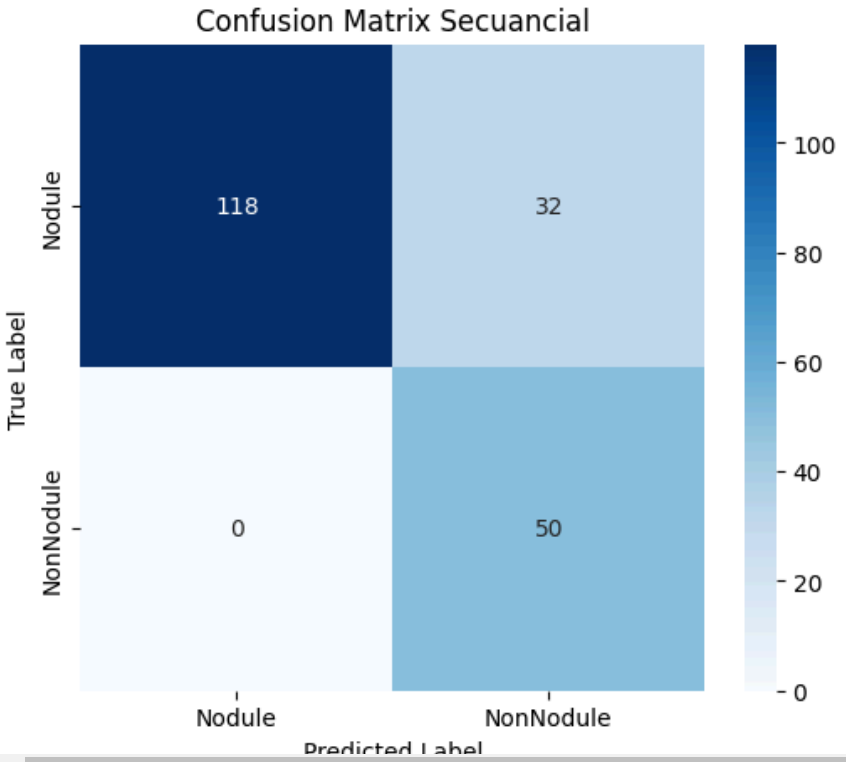
```
cm = np.array([[TP, FN], [FP, TN]])
print(cm)
```

```
[[118  32]
 [  0  50]]
```

```
accuracy = (TP + TN) / (TP + TN + FP + FN)
precision = TP / (TP + FP) if (TP + FP) != 0 else 0
recall = TP / (TP + FN) if (TP + FN) != 0 else 0
f1_score = 2 * (precision * recall) / (precision + recall) if (precision + recall) != 0 else 0

print(f"\nAccuracy: {accuracy:.2f}")
print(f"Precision: {precision:.2f}")
print(f"Recall: {recall:.2f}")
print(f"F1 Score: {f1_score:.2f}\n")
```

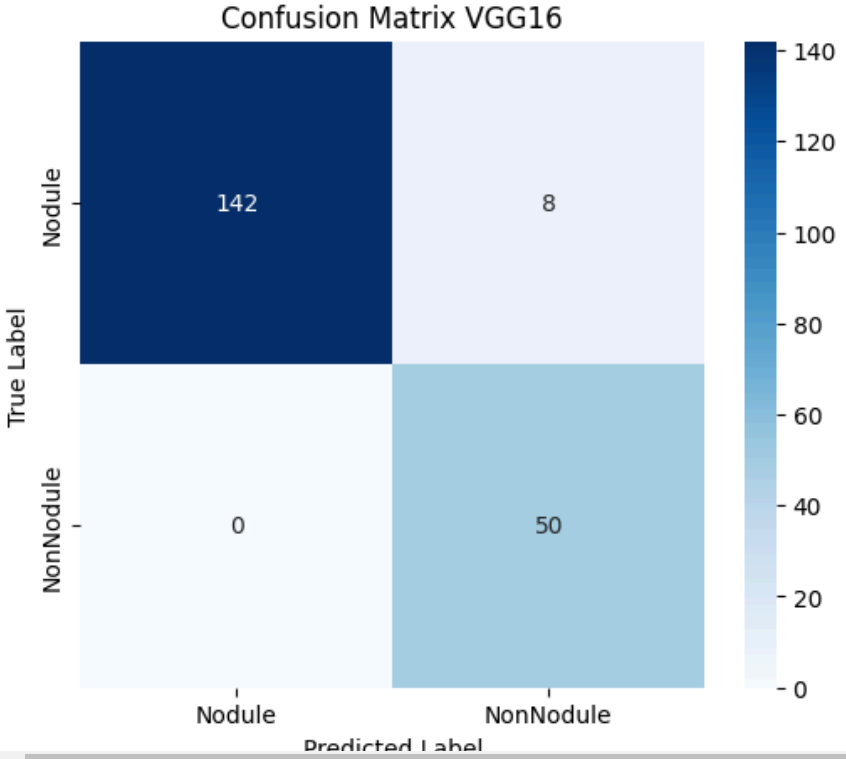
```
# Create a heatmap of the confusion matrix
plt.figure(figsize=(6, 5))
# Replace val_generator.class_indices.keys() with your actual class labels
class_labels = ['Nodule', 'NonNodule'] # Update this with your actual class names
sns.heatmap(cm, annot=True, fmt="d", cmap="Blues", xticklabels=class_labels, yticklabels=class_labels)
plt.ylabel('True Label')
plt.xlabel('Predicted Label')
plt.title('Confusion Matrix Secuancial')
plt.show()
```

```
Accuracy: 0.84
Precision: 1.00
Recall: 0.79
F1 Score: 0.88
```



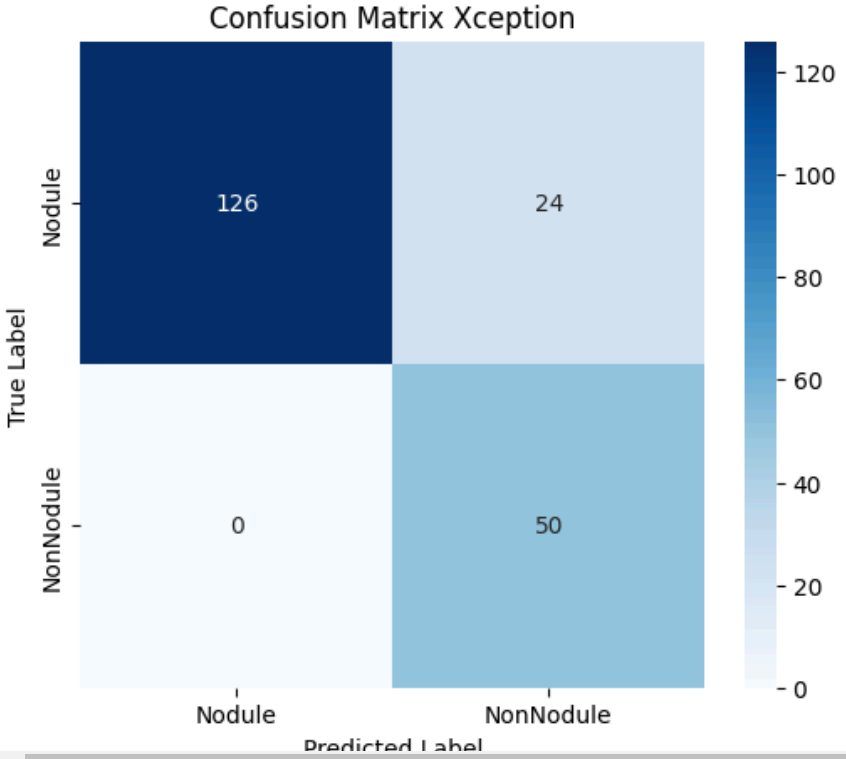

```
accuracy = (TP + TN) / (TP + TN + FP + FN)
precision = TP / (TP + FP) if (TP + FP) != 0 else 0
recall = TP / (TP + FN) if (TP + FN) != 0 else 0
f1_score = 2 * (precision * recall) / (precision + recall) if (precision + recall) != 0 else 0

print(f"\nAccuracy: {accuracy:.2f}")
print(f"Precision: {precision:.2f}")
print(f"Recall: {recall:.2f}")
print(f"F1 Score: {f1_score:.2f}\n")

plt.figure(figsize=(6, 5))
class_labels = ['Nodule', 'NonNodule']
sns.heatmap(cm, annot=True, fmt="d", cmap="Blues", xticklabels=class_labels, yticklabels=class_labels)
plt.ylabel('True Label')
plt.xlabel('Predicted Label')
plt.title('Confusion Matrix VGG16')
plt.show()
```

```
Accuracy: 0.96
Precision: 1.00
Recall: 0.95
F1 Score: 0.97
```




```
accuracy = (TP + TN) / (TP + TN + FP + FN)
precision = TP / (TP + FP) if (TP + FP) != 0 else 0
recall = TP / (TP + FN) if (TP + FN) != 0 else 0
f1_score = 2 * (precision * recall) / (precision + recall) if (precision + recall) != 0 else 0

print(f"\nAccuracy: {accuracy:.2f}")
print(f"Precision: {precision:.2f}")
print(f"Recall: {recall:.2f}")
print(f"F1 Score: {f1_score:.2f}\n")

# Create a heatmap of the confusion matrix
plt.figure(figsize=(6, 5))
class_labels = ['Nodule', 'NonNodule']
sns.heatmap(cm, annot=True, fmt="d", cmap="Blues", xticklabels=class_labels, yticklabels=class_labels)
plt.ylabel('True Label')
plt.xlabel('Predicted Label')
plt.title('Confusion Matrix Xception')
plt.show()
```

```
Accuracy: 0.88
Precision: 1.00
Recall: 0.84
F1 Score: 0.91
```



Grad-Cam

```
class GradCam():

  def __init__(self, model, last_conv_layer_name):

        self.model = model
        self.last_conv_layer_name = last_conv_layer_name

        # get the layer object of the last convolutionl layer
        last_conv_layer = self.model.get_layer(self.last_conv_layer_name).output

        # create a model that maps the input image to the activations
        # of the last conv layer as well as the output predictions
        self.grad_model = tf.keras.models.Model(
            [model.inputs], [model.get_layer(last_conv_layer_name).output, model.output])



  def compute_heatmap(self, img_array, pred_index=None):

        '''
        Purpose:
            Get the class level activations heatmap for the CNN network. The activation maps help us
            identify & localize the regions, patterns of the image the neural network looks at, and
            activates the region around the patterns. To visualize the activation maps, this function
            works on the output of the LAST CONVOLUTIONAL LAYER and the final CLASSIFICATION LAYERS.

        Parameters:
            1. img_array - the array representation of the image for which the activation maps are to be
                           visualized.
```

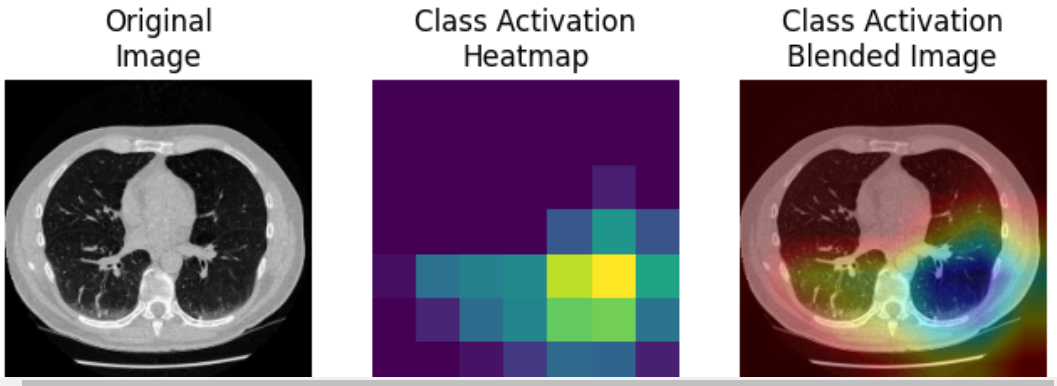
```
            2. model - the CNN model whose activations are to be analyzed.
            3. last_conv_layer_name - name of the last convolutional layer of the model.
            4. pred_index (optional): The index of the largest value in the tensor.
                                      The unit which contributed the most towards the predicted result.

        Return Value:
            heatmap - The heatmap showing the more active regions the CNN looked at in deciding the class
                      for the image.

        '''

        # Compute the gradient of the top predicted class for our input image
        # with respect to the activations of the last conv layer
        with tf.GradientTape() as tape:
            last_conv_layer_output, preds = self.grad_model(img_array)
            tensor_preds = preds[0]
            if pred_index is None:
                pred_index = tf.argmax(tensor_preds[0]) # returns the index of the largest value in the ter
            class_channel = tensor_preds[:, pred_index]

            # last_conv_layer_output, preds = self.grad_model(img_array)
            # if pred_index is None:
            #     pred_index = tf.argmax(preds[0]) # returns the index of the largest value in the tensor.
            # class_channel = preds[:, pred_index]

        # return [last_conv_layer_output]

        # This is the gradient of the output neuron (top predicted or chosen)
        # with regard to the output feature map of the last conv layer
        grads = tape.gradient(class_channel, last_conv_layer_output)

        # This is a vector where each entry is the mean intensity of the gradient
        # over a specific feature map channel.
        # get the mean of elements across dimensions of a tensor.
        pooled_grads = tf.reduce_mean(grads, axis=(0, 1, 2))

        # We multiply each channel in the feature map array
        # by "how important this channel is" with regard to the top predicted class
        # then sum all the channels to obtain the heatmap class activation
        last_conv_layer_output = last_conv_layer_output[0]
        heatmap = last_conv_layer_output @ pooled_grads[..., tf.newaxis]
        heatmap = tf.squeeze(heatmap)

        # For visualization purpose, we will also normalize the heatmap between 0 & 1
        heatmap = tf.maximum(heatmap, 0) / tf.math.reduce_max(heatmap)

        return heatmap.numpy()

    def GetSuperImposedCAMImage(self, heatmap, image, alpha = 0.5):

        ''''
        Purpose:
            Get the super imposed or the blended version of the image comprising of the class activations
            heatmap and the original image.

        Parameters:
            1. heatmap - The class activation heatmap for the concerned image.
            2. image - The original image for which the activation image is being calculated.
            3. alpha (optional) - A number between 0 & 1 to control the weighted blending of the images.

        Return Value:
            superImposedImage - The blended version of the original image and the corresponding class
                                activation map.

        '''
```

```
        # Rescale the image to 0-255
        image = image * 255

        # Rescale heatmap to 0-255
        # Resize to the size of the image
        # Apply colormap
        heatmap = np.uint8(255 * heatmap)

        # upsample the class actiavtion to match the size of the original image
        # before the 2 images can be blended for final visualization.
        heatmap = cv2.resize(heatmap, (image.shape[1], image.shape[0]))

        heatmap = cv2.applyColorMap(heatmap, cv2.COLORMAP_JET)

        # Superimpose the heatmap on original image
        superImposedImage = heatmap * alpha + image
        superImposedImage = tf.keras.preprocessing.image.array_to_img(superImposedImage)

        return superImposedImage


    def DisplaySuperImposedImages(self, image, heatmap,superimposed_img):

        '''
        Purpose:
            Display the original image and the corresponding class activation blended image.

        Parameters:
            1. image - The original image for which the activation image is being calculated.
            2. superimposed_img - The blended version of the original image and the corresponding class
                                  activation map.

        Return Value:
            NONE

        '''

        fig, ax = plt.subplots(1, 3, figsize=(8, 12))

        ax[0].imshow(image)
        ax[1].imshow(heatmap)
        ax[2].imshow(superimposed_img)

        ax[0].title.set_text('Original\nImage')
        ax[1].title.set_text('Class Activation\nHeatmap')
        ax[2].title.set_text('Class Activation\nBlended Image')

        ax[0].axis('off')
        ax[1].axis('off')
        ax[2].axis('off')
```

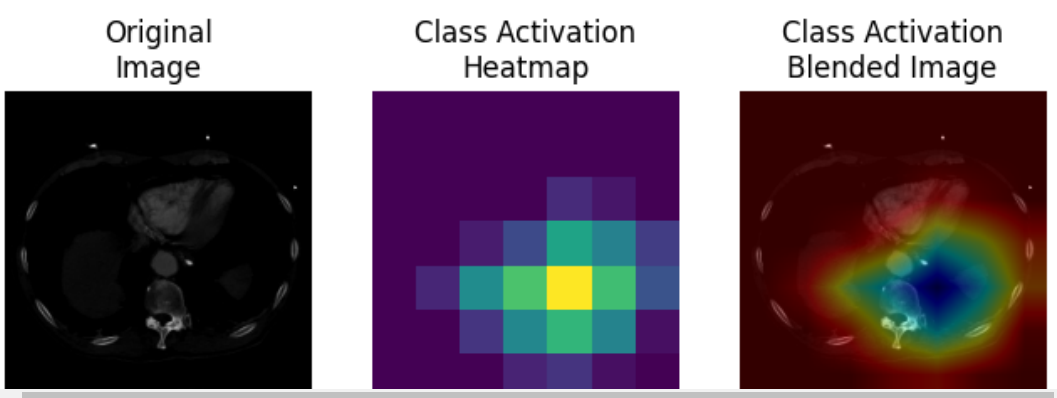
```
imagen = f'/content/test/images/{test_data[0]}'
load_image = cv2.imread(imagen)
img = cv2.cvtColor(load_image, cv2.COLOR_BGR2RGB)
img = cv2.resize(img, (224, 224))
img = cv2.normalize(img, None, 0, 1.0, cv2.NORM_MINMAX, dtype = cv2.CV_32F)
```

G-CAM Xception

```
model = tf.keras.models.load_model('modeloXce.keras')


gCamXception = GradCam(model, "block14_sepconv2")


# reshape img to a tensor of shape (1, 128, 128, 3)
imgAsTensor = img.reshape((1, img.shape[0], img.shape[1], img.shape[2]))

# generate the heatmap
heatmap = gCamXception.compute_heatmap(imgAsTensor)

superImposedImg = gCamXception.GetSuperImposedCAMImage(heatmap = heatmap, image = img)
gCamXception.DisplaySuperImposedImages(image = img, heatmap = heatmap, superimposed_img = superImposedImg)
```

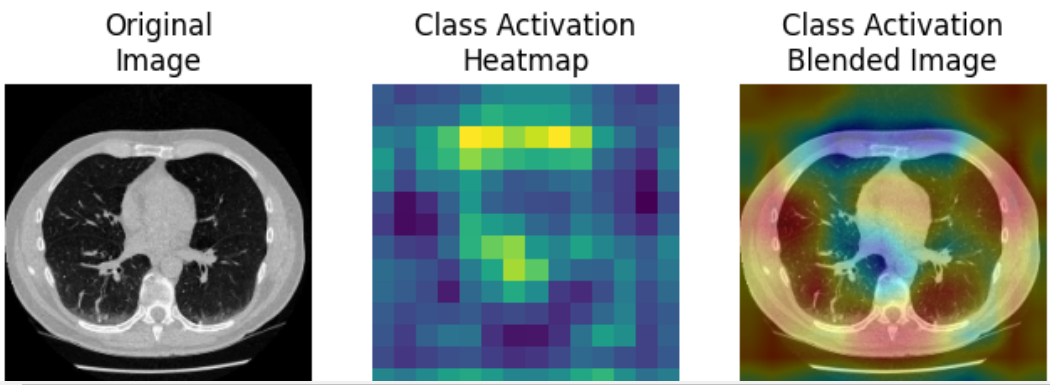



```
# generate the heatmap
heatmap = gCamXception.compute_heatmap(imgAsTensor)

superImposedImg = gCamXception.GetSuperImposedCAMImage(heatmap = heatmap, image = img)
gCamXception.DisplaySuperImposedImages(image = img, heatmap = heatmap, superimposed_img = superImposedImg)
```



G-CAM VGG

```
model = tf.keras.models.load_model('modeloVGG.keras')


gCamVgg = GradCam(model, "block5_conv3")


imgAsTensor = img.reshape((1, img.shape[0], img.shape[1], img.shape[2]))

# generate the heatmap
heatmap = gCamVgg.compute_heatmap(imgAsTensor)

superImposedImg = gCamVgg.GetSuperImposedCAMImage(heatmap = heatmap, image = img)
gCamVgg.DisplaySuperImposedImages(image = img, heatmap = heatmap, superimposed_img = superImposedImg)
```

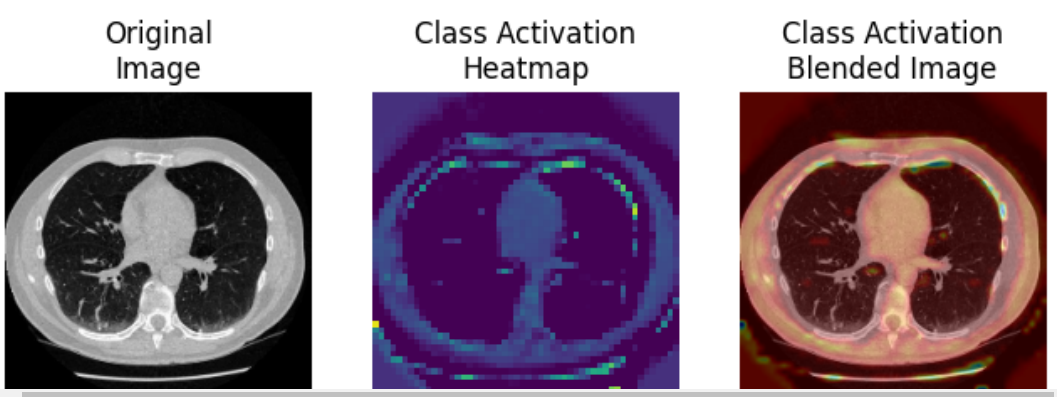


G-CAM Secuencial

```
model = tf.keras.models.load_model('modelo.keras')
```

```
gCamSeq = GradCam(model, "conv2d_8")
```

```
imgAsTensor = img.reshape((1, img.shape[0], img.shape[1], img.shape[2]))

# generate the heatmap
heatmap = gCamSeq.compute_heatmap(imgAsTensor)

superImposedImg = gCamSeq.GetSuperImposedCAMImage(heatmap = heatmap, image = img)
gCamSeq.DisplaySuperImposedImages(image = img, heatmap = heatmap, superimposed_img = superImposedImg)
```



Cálculo IoU

Crear subconjunto para las pruebas

Definir subconjunto de datos a dentro del de test en el que se conocen las anotacoines correspondientes con las localizaciones

```
def txt_yolo_check(file_path):

  with open(file_path, 'r') as file:
    data = file.read()
    labels = data.split()
    if len(labels) > 2:
      return True
    else:
```

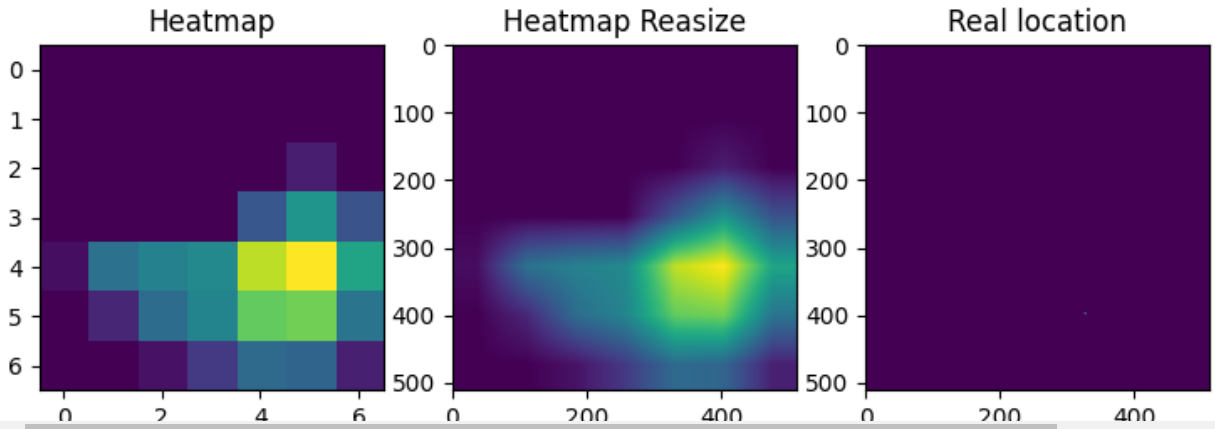
```
        return False


test_labels = os.listdir('/content/test/labels/')
total_iou_data = []

for label in test_labels:
  if txt_yolo_check(f'/content/test/labels/{label}'):
    total_iou_data.append(label)

print(len(total_iou_data))
```

```
120
```

Funciones

```
from google.colab.patches import cv2_imshow

def read_txt(file_path):
    with open(file_path, 'r') as file:
        lines = file.readlines()
        result = []
        for line in lines:
          labels = line.split()
          coordenates = [float(labels[1]), float(labels[2]), float(labels[3]), float(labels[4])]
          result.append(coordenates)
    return result


# Función para convertir formato YOLO a una máscara binaria
def yolo_to_mask(yolo_coords, img_shape):
    """
    Convierte coordenadas YOLO a una máscara binaria.
    :param yolo_coords: Lista [class_id, x_center, y_center, width, height] en formato YOLO.
    :param img_shape: Tamaño de la imagen original (altura, ancho).
    :return: Máscara binaria con el bounding box.
    """
    # Extraer las dimensiones de la imagen
    img_height, img_width = img_shape[0], img_shape[1]

    # Desempaquetar las coordenadas del formato YOLO
    # class_id, x_center, y_center, width, height = yolo_coords
    x_center, y_center, width, height = yolo_coords

    # Convertir coordenadas normalizadas a coordenadas en píxeles
    x_center = int(x_center * img_width)
    y_center = int(y_center * img_height)
    box_width = int(width * img_width)
    box_height = int(height * img_height)

    # Calcular las esquinas del bounding box
    x_min = max(0, x_center - box_width // 2)
    x_max = min(img_width, x_center + box_width // 2)
    y_min = max(0, y_center - box_height // 2)
    y_max = min(img_height, y_center + box_height // 2)

    # Crear una máscara binaria vacía
    mask = np.zeros((img_height, img_width), dtype=np.uint8)

    # Dibujar el bounding box como una región rectangular en la máscara
    mask[y_min:y_max, x_min:x_max] = 1

    return mask
```

```
def binarize_grad_cam(grad_cam, threshold=0.5):
    binary_grad_cam = grad_cam > threshold
    return binary_grad_cam.astype(np.uint8)


def calculate_iou(mask1, mask2):
    # Calculamos la intersección y la unión
    intersection = np.logical_and(mask1, mask2)
    union = np.logical_or(mask1, mask2)

    # La IoU es la suma de la intersección dividida por la suma de la unión
    iou = np.sum(intersection) / np.sum(union)

    iou = math.ceil(iou)
    return iou


# Tamaño de la imagen original (altura, ancho)
img_shape = (512, 512)

yolo_nodules = read_txt(f'/content/test/labels/{test_data[0].replace(".jpg", ".txt")}')
ground_truth_mask = yolo_to_mask(yolo_nodules[0], img_shape)

# Supongamos que grad_cam_output es la matriz generada por Grad-CAM con forma (14x14)
heatmap_resized = cv2.resize(heatmap, (512, 512))

# Normalizamos para que los valores de grad_cam_resized estén en [0, 1]
heatmap_resized = (heatmap_resized - np.min(heatmap_resized)) / (np.max(heatmap_resized) - np.min(heatmap_re

plt.figure(figsize=(9, 3))

plt.subplot(1, 3, 1)
plt.imshow(heatmap, cmap='viridis')
plt.title("Heatmap")

plt.subplot(1, 3, 2)
plt.imshow(heatmap_resized, cmap='viridis') # ground_truth_mask * 255
plt.title("Heatmap Reasize")

plt.subplot(1, 3, 3)
plt.imshow(ground_truth_mask, cmap='viridis')
plt.title("Real location")

plt.show()
```




```
binary_grad_cam = binarize_grad_cam(heatmap_resized, threshold=0.5)
iou = calculate_iou(binary_grad_cam, ground_truth_mask)
print(iou)
```

1

Proceso y resultados

```
img_shape = (512, 512)
iou_result = []

for label in total_iou_data:
  # Load the image
  load_image = cv2.imread(f'/content/test/images/{label.replace("txt", "jpg")}')
  img = cv2.cvtColor(load_image, cv2.COLOR_BGR2RGB)
  img = cv2.resize(img, (224, 224))
  img = cv2.normalize(img, None, 0, 1.0, cv2.NORM_MINMAX, dtype = cv2.CV_32F)

  # reshape img to a tensor of shape (1, 128, 128, 3)
  imgAsTensor = img.reshape((1, img.shape[0], img.shape[1], img.shape[2]))

  # generate the heatmap
  # heatmap = gCamXception.compute_heatmap(imgAsTensor)
  heatmap = gCamVgg.compute_heatmap(imgAsTensor)

  # Supongamos que grad_cam_output es la matriz generada por Grad-CAM con forma (14x14)
  heatmap_resized = cv2.resize(heatmap, (512, 512))
  # Normalizamos para que los valores de grad_cam_resized estén en [0, 1]
  heatmap_resized = (heatmap_resized - np.min(heatmap_resized)) / (np.max(heatmap_resized) - np.min(heatmap

  real_nodules = read_txt(f'/content/test/labels/{label}')

  for real_nodule in real_nodules:

    ground_truth_mask = yolo_to_mask(real_nodule, img_shape)
    binary_grad_cam = binarize_grad_cam(heatmap_resized, threshold=0.5)
    iou = calculate_iou(binary_grad_cam, ground_truth_mask)

    iou_result.append(iou)
```